

D:\llm-projects\projects\study-notes-manager\docs\tasks.md

Build **the backend CRUD API first**. Don't start with LangChain/LangGraph or the frontend.

For the **AI Study Notes Manager**, follow this order:

Step 1 — Create the FastAPI project

Start with:

```
backend/
├── app/
│   ├── main.py
│   ├── api/
│   │   └── notes.py
│   ├── schemas/
│   │   └── note.py
│   ├── models/
│   │   └── note.py
│   └── db/
│       └── database.py
└── pyproject.toml
```

First make FastAPI run:

```
GET /
→ {"message": "API is running"}
```

Step 2 — Build CRUD without AI

Create the five endpoints:

```
POST    /notes
GET     /notes
GET     /notes/{id}
PUT     /notes/{id}
DELETE  /notes/{id}
```

Your first goal is:

```
Create Note
    ↓
Database
    ↓
```

Get Note



Update Note



Delete Note

Test everything in **FastAPI Swagger /docs**.

Step 3 — Connect PostgreSQL

Create the notes table:

```
notes
-----
id
title
subject
content
created_at
updated_at
```

At this point you should have a completely working **Notes REST API**.

Step 4 — Build Next.js frontend

Only after the API works.

Build:

Dashboard



Notes List



Create Note



View Note



Edit Note



Delete Note

Connect:

Next.js
↓
FastAPI
↓
PostgreSQL

Now you have a real full-stack CRUD application.

Step 5 — Add LangChain

Then add one AI feature first:

POST /notes/{id}/summarize

Flow:

Note
↓
FastAPI
↓
LangChain
↓
LLM
↓
Summary
↓
Next.js

Don't add five AI features at once.

Step 6 — Add LangGraph

After LangChain works, create a small graph:

START
↓
Analyze Note
↓
Generate Summary
↓
Extract Key Concepts
↓
Generate Quiz

↓
END

Your actual roadmap

1. FastAPI setup
↓
2. FastAPI CRUD
↓
3. PostgreSQL
↓
4. Test API
↓
5. Next.js UI
↓
6. Connect frontend + backend
↓
7. LangChain
↓
8. LangGraph
↓
9. Authentication
↓
10. RAG / PDF upload / streaming

So your first coding task should be:

Create a FastAPI project and implement `POST /notes` and `GET /notes` using an in-memory Python list first.

Once that works, move to PostgreSQL. This keeps the project simple and lets you understand each layer instead of debugging Next.js + FastAPI + database + LangGraph all at once.